CSW323-Operating Systems-2

Introduction

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Course Descriptions

- **Prerequisites:** CSW 242 Operating Systems (1)

- **Assessment:**
  - Course Work (Assignments , Labs, Projects): 20%
  - Midterm Exam: 20%
  - Final Examination: 60%

- **Text Book(s):**
  - Operating System Concepts – 8th Edition, Silberschatz, Galvin and Gagne ©2009.

Outline

- What is an Operating System?
- Computer System Structure
- Computer-System Operation
- Operating System Functions
- Operating Systems Types
- Storage Structure
- Multiprogramming
- Process Concept
- Process Scheduling

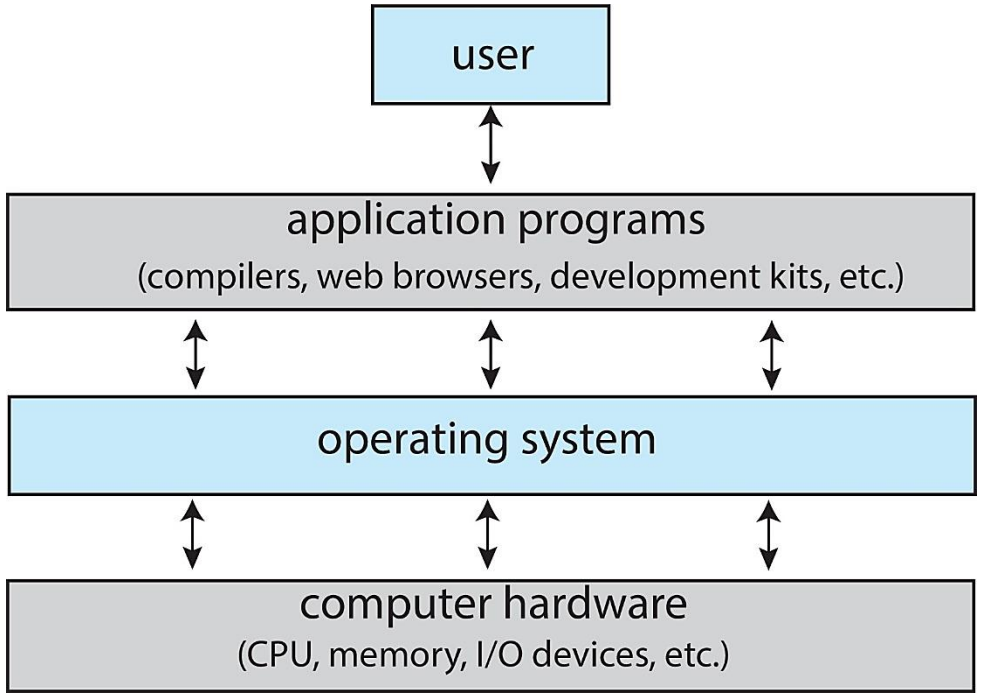
What is an Operating System?

- A program that acts as an interface between a user of a computer and the computer hardware.
- **Operating system goals:**
  - Execute user programs and make solving user problems easier.
  - Make the computer system convenient to use.
  - Use the computer hardware in an efficient manner.

Computer System Structure

- Computer system can be divided into four components:
  - **Hardware** – provides basic computing resources
    - CPU, memory, I/O devices.
  - **Operating system**
    - Controls and coordinates use of hardware among various applications and users.
  - **Application programs** – define the ways in which the system resources are used to solve the computing problems of the users
    - Word processors, compilers, web browsers, database systems, video games.
  - **Users**
    - People, machines, other computers.

Computer System Structure



Computer-System Operation

- I/O devices and the CPU can execute concurrently.
- Each device controller is in charge of a particular device type.
- Each device controller has a local buffer.
- Each device controller type has an operating system device driver to manage it
- CPU moves data from/to main memory to/from local buffers
- I/O is from the device to local buffer of controller
- Device controller informs CPU that it has finished its operation by causing an interrupt.

Computer-System Operation

- Interrupt transfers control to the interrupt service routine generally, through the interrupt vector, which contains the addresses of all the service routines.
- Interrupt architecture must save the address of the interrupted instruction.
- **A trap** or **exception** is a software-generated interrupt caused either by an error or a user request.
- An operating system is interrupt driven.

Operating System Functions

- Activities performed by the operating system:
  - Perform common computer hardware functions.
  - Provide a user interface and input/output management.
  - Manage system memory.
  - Manage processing tasks.
  - Provide networking capability.
  - Control access to system resources.
  - Manage files.

Operating Systems Types

- **Embedded OS:** is a specialized operating system designed to perform a specific task for a device that is not a computer.
- **Mobile OS:** Android, iOS, Ubuntu Touch, Windows Touch.
- **RTOS:** A real-time operating system is a software component that rapidly switches between tasks, giving the impression that multiple programs are being executed at the same time on a single processing core. Ex: QNX, VxWorks, RTLinux.
- **Secure Environments:** SeLinux, SeL4.
- **Servers:** Redhat, Ubuntu, Windows Server.
- **Desktops:** Mac OS, Windows, Ubuntu(With a built-in firewall and virus protection software).

Storage Structure

- **Main memory** – only large storage media that the CPU can access directly
  - Random access
  - Typically volatile
  - Typically random-access memory in the form of Dynamic Random-Access Memory (DRAM).
- **Secondary storage** – extension of main memory that provides large nonvolatile storage capacity.

Storage Structure

- **Hard Disk Drives (HDD)** – rigid metal or glass platters covered with magnetic recording material
- Disk surface is logically divided into tracks, which are subdivided into sectors.
- Non-volatile memory (NVM) devices– faster than hard disks, nonvolatile
- Various technologies
- Becoming more popular as capacity and performance increases, price drops.

Storage Structure

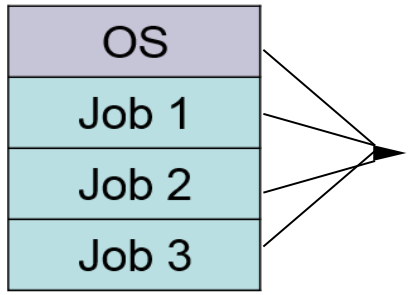
- Storage systems organized in hierarchy:
  - Speed
  - Cost
  - Volatility
- **Caching** – copying information into faster storage system; main memory can be viewed as a cache for secondary storage.

Caching

- Important principle, performed at many levels in a computer (in hardware, operating system, software)
- Information in use copied from slower to faster storage temporarily
- Faster storage (cache) checked first to determine if information is there
  - If it is, information used directly from the cache (fast)
  - If not, data copied to cache and used there
- Cache smaller than storage being cached
  - Cache management important design problem
  - Cache size and replacement policy

Multiprogramming

- Multiple jobs in memory:
  - When one waits for I/O the next job executes
- OS controls:
  - Scheduling of jobs
- Protection between jobs



Memory partitions

Multiprogramming with 3 jobs in memory

Process Concept

- An operating system executes a variety of programs: Batch system – jobs
- Textbook uses the terms job and process almost interchangeably.
- **Process** – a program in execution; process execution must progress in sequential fashion.

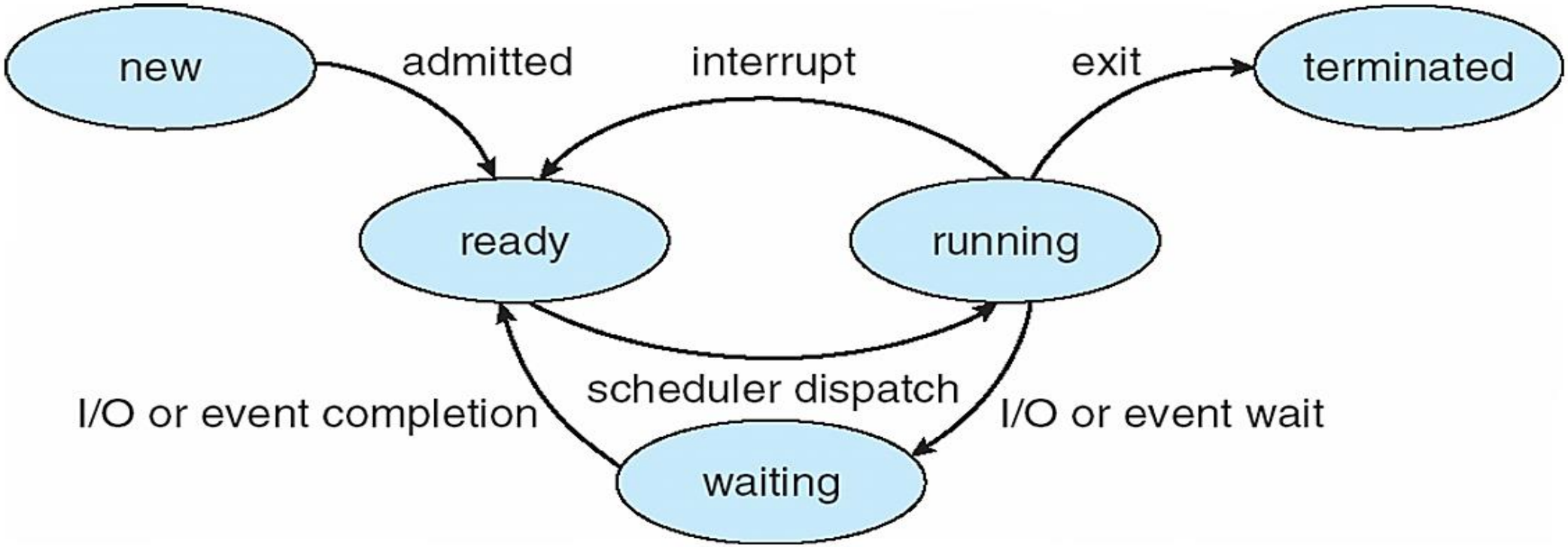
The Process

- Multiple parts:
  - The program code, also called **text section**
  - Current activity including **program counter**, processor registers
  - **Stack** containing temporary data
  - **Data section** containing global variables
  - **Heap** containing memory dynamically allocated during run time

Process State

- As a process executes, it changes state
  - new: The process is being created
  - running: Instructions are being executed
  - waiting: The process is waiting for some event to occur
  - ready: The process is waiting to be assigned to a processor
  - terminated: The process has finished execution

Diagram of Process State



Process Control Block (PCB)

Information associated with each process:

- Process state
- Program counter
- CPU registers
- CPU scheduling information
- Memory-management information
- Accounting information
- I/O status information

Process Scheduling

- Process scheduler selects among available processes for next execution on CPU.
- Maintains scheduling queues of processes.
- **Job queue** – set of all processes in the system.
- **Ready queue** – set of all processes residing in main memory, ready and waiting to execute.
- **Device queues** – set of processes waiting for an I/O device.
- Processes migrate among the various queues.

Schedulers

- **Long-term scheduler** (or job scheduler) – selects which processes should be brought into the ready queue.
- **Short-term scheduler** (or CPU scheduler) – selects which process should be executed next and allocates CPU.